

LESSON 14.1

DETERMINING SAMPLE SPACE USING LISTS AND TREE DIAGRAMS



LESSON INTRODUCTION

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

LEARNING TARGETS

- I can determine the sample space of an experiment using a list.
- I can understand how a tree diagram can be used to determine the sample space of an experiment.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.1 Determine the sample space for a simple experiment.

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

WORKING TOWARD

MA.912.DP.4.1 Describe events as subsets of a sample space using characteristics, or categories, of the outcomes; or as unions, intersections or complements of other events.

MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.

ESSENTIAL QUESTIONS

- How can the sample space of a repeated random process be determined?
- How are sample space lists and tree diagrams the same? How are they different?

LESSON NARRATIVE

This lesson activates prior knowledge regarding probability to determine the sample space of random processes. Students will build on this skill by exploring the number of outcomes in repeated experiments as well. Then, students learn to create tree diagrams as a visual representation of the sample space. The primary learning outcome in this lesson focuses on identifying the sample space of an experiment using lists and tree diagrams.

COMPONENT SUMMARY

14.1.1 WARM-UP (~5 MIN.)

Students convert numbers between fractions, decimals, and percentages

14.1.3 EXPLORATION (10-15 MIN.)

Explore sample spaces of repeated experiments

14.1.5 WRAP-UP (~5 MIN.)

Students reflect on their understanding of determining sample space

14.1.2 REFRESHER (5-7 MIN.)

Revisit how to determine the sample space and calculate simple probability

14.1.4 GUIDED INSTRUCTION (15-18 MIN.)

Introduction to tree diagrams

RESOURCES

GLOSSARY

Key Terms:

- Outcome
- Sample space
- Event
- Repeated experiment
- Tree diagram

Additional Terms:

- N/A

MATERIALS

- (Optional) Dice
- (Optional) Coins
- (Optional) Highlighters

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may interpret theoretical probability as “guaranteed.” (i.e. since $P(H) = \frac{1}{2}$, then we are guaranteed for one flip to result in heads out of two flips).
- Students may misinterpret the events in the sample space as being the same. (i.e. HT vs. TH).
- Students may not identify the tree diagram as a sample space, and instead think that they need to list the sample space after creating a tree diagram.

THINGS TO CONSIDER

- Determine if your students need the support of all or part of the refresher.
- Equivalent forms will be heavily used in this unit. Consider using 14.1.1 Warm-Up as a formative assessment to determine whether additional support for expressions with rational numbers is needed prior to moving on to subsequent lessons in the unit.
- If the optional coins aren’t available, students can use an online simulator tool to run simulations of flipping coins.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 14.1.4 Guided Instruction, consider providing an opportunity for students to utilize this language routine after introducing question 3 before the Wrap-Up. Allow each student to reflect on tree diagrams and discuss the provided prompts with their partner as they process what they have learned thus far in the lesson. Creating a low-stakes opportunity for students to discuss their ideas will build confidence and allow them to gather thoughts before sharing with the whole group.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.2.1 Multiple Representations**

Students build understanding of sample spaces by creating lists and tree diagrams throughout the lesson. Encourage students to compare the two representations in the 14.1.4 Guided Instruction as they develop an understanding of methods used to compare sample spaces. Discuss how both representations can be used for sample spaces, comparing similarities and differences, as well as the benefits of the process of creating either.

DIFFERENTIATE INSTRUCTION

14.1.3 Exploration is text and reading heavy. As students work through this section with their partner, encourage them to use a highlighter to highlight phrases and words to call attention to keywords.

Because this topic is text-heavy, “Mark-Up Text” is a good activity to help ELL students begin to feel comfortable translating words into mathematical expressions.

- Provide students with the following symbols or allow students to create their own annotations.
 - ✓ - background knowledge, they know what to do.
 - ? - confusing information; is this needed to solve the problem?
 - ! - marks where a variable is needed to represent a number
 - + - extra info in word problem; not needed.
- Using a small piece of text, model for students how you would Mark Up before releasing them to do the same.
- Students will read the text slowly and carefully, focusing on one sentence at a time and marking text based on the key. Then they will turn to a partner to discuss each sentence and compare where they made marks.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

14.1.1 WARM-UP: EQUIVALENT FORMS**DIFFERENTIATE INSTRUCTION**

After students have completed the Warm-Up independently, consider having students compare their work with their partner, discuss what they struggled with, and resolve any differences.

DIFFERENTIATE INSTRUCTION

The results of this Warm-Up can be used to identify students who may need additional support when expressing probabilities as fractions, decimals, and percentages in this unit.

**14.1.1 Warm-Up: Equivalent Forms**

1. Complete the table below so each row shows the same value in three equivalent forms.

Fraction (In Simplest Form)	Decimal	Percentage (To the Nearest Tenth)
$\frac{7}{12}$	0.58 $\bar{3}$	58.3%
$\frac{3}{5}$	0.6	60.0%
$\frac{3}{4}$	0.75	75.0%
$\frac{7}{20}$	0.35	35.0%

**14.1.2 Refresher: Determining the Sample Space**

Math is used every day to help make decisions. One branch of math that can be used to inform decisions is probability. In order to study the probability of a random process, it is important to be able to identify the sample space of the process.

An **outcome** is a possible result of an experiment.



In a probability model for a random process, the **sample space** is a list of the individual outcomes that are to be considered.



An **event** is a set of possible outcomes that can result from an experiment. In general, an event is any subset of a sample space.

1. A familiar random process is rolling a fair die. List the possible outcomes of rolling a fair die.
{1,2,3,4,5,6}

This list is the **sample space** for the random process of rolling a fair die.

A sample space is useful for finding probabilities when each outcome in a random process is equally likely.

2. Let A be the event, "rolling an even number."
 - a. List the elements in Event A .
{2,4,6}
 - b. Find $P(A)$.

$$P(A) = \frac{\text{number of outcomes in event } A}{\text{total number of outcomes in the sample space}} = \frac{3}{6}$$

- c. Write $P(A)$ as a decimal and as a percentage.
.50; 50%
3. A fair coin is flipped.
 - a. What is the sample space?
{heads, tails}
 - b. What is the probability of landing on heads?
 $\frac{1}{2}$

14.1.2 REFRESHER: DETERMINING THE SAMPLE SPACE**DIFFERENTIATE INSTRUCTION**

Determine if your students need the support of all or part of the Refresher.

ENRICHMENT

To ensure that students are able to articulate what the vocabulary terms mean in their own words. Ask students questions like:

- What does it mean for a process to be random?
- How do sample spaces relate to events?

TEACHER MOVES

This component should be guided by the teacher to ensure a quick refresher of these concepts from the prior grade. If deemed unnecessary based on students' recollection of these concepts, question 3 could be skipped.

14.1.3 EXPLORATION: REPEATED COIN FLIPPING

TEACHER MOVES

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to during the whole-group debrief at the end. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

DIFFERENTIATE INSTRUCTION

Mark-Up Text

This component is text heavy, so struggling readers and ELL students can benefit from reading the text slowly and carefully to make note of the important information. Encouraging students to highlight, label, and annotate as they read through this problem will help ensure students understand what is happening in the problem before they engage with any questions.



14.1.3 Exploration: Repeated Coin Flipping

Consider the random process of flipping *two* fair coins and getting all heads.

1. Examine the work below with your partner and decide whom you agree with.

Rhaheim and Octavia are discussing their thoughts on how likely it is to get only heads when two fair coins are flipped.

Octavia says, "The likelihood of getting all heads when two coins are flipped is $\frac{1}{2}$. Since half of the outcomes when one coin is flipped are heads, then half of the outcomes when two coins are flipped will also be heads."

Rhaheim says, "I'm not exactly sure what the likelihood of getting all heads when flipping two coins is, but I know it's not $\frac{1}{2}$. When two coins are flipped, if the first coin is heads, the second coin could be heads or tails. In only these two outcomes, HH, and HT, the likelihood of all heads is $\frac{1}{2}$, but there are more outcomes, so this likelihood will change. I'm just not sure how many more outcomes there are."

Work with your partner to decide if you agree more with Rhaheim or with Octavia and why. Write your explanation.

Answers will vary. We think Rhaheim is correct because flipping two coins changes the possible outcomes. You can also get TT and TH. The first coin could be tails and the second coin could be heads or tails as well.

2. List the sample space of flipping a fair coin twice. On the first toss, the possible options are heads and tails. On the second toss, the two options are heads and tails.

$\{HH, HT, TH, TT\}$

3. Use the sample space to determine how likely it is to flip all heads when two coins are flipped. Create a ratio to compare the number of outcomes with only heads to the total number of outcomes in the sample space.

$\frac{1}{4}$

4. What if *three* fair coins are flipped? How will the sample space and the likelihood of flipping only heads change?

- a. Make a prediction below.

Answers will vary. There will still be one chance out of the total outcomes to flip only heads.

- b. Work with your partner to determine the sample space when three fair coins are flipped.

$\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

- c. Find the likelihood of flipping all heads when three fair coins are flipped.

$\frac{1}{8}$

- d. Compare your work with that of another pair of students. Revise your work to make it more accurate, if necessary.

Answers will vary.

5. Work as a group of four to investigate how the sample space and the likelihood of flipping all heads changes when four fair coins are flipped. Fill in the table below with the information for flipping 1, 2, and 3 coins.

Number of Coins Flipped	Number of Outcomes in the Sample Space	Likelihood of Getting All Heads
1	2	$\frac{1}{2}$
2	4	$\frac{1}{4}$
3	8	$\frac{1}{8}$
4	16	$\frac{1}{16}$

6. Use the table above to make a prediction about how the sample space and likelihood of getting all heads will change when *four* fair coins are flipped.

Answers will vary. We noticed a pattern that each time we added another coin, the number of outcomes doubled but the chance of getting all heads was still 1 out of the total outcomes. We noticed that when you flip 2 coins there are 4 outcomes or 2^2 , and when you flip 3 coins the number of outcomes is 2^3 , so following the pattern, flipping 4 coins would be 2^4 or 16 outcomes.

Flipping a fair coin multiple times is an example of a **repeated experiment**. Each repetition — in this case, a coin toss — is a trial.



A **repeated experiment** is a random experiment completed again with the same conditions and parameters as a previous one.

14.1.3 EXPLORATION: REPEATED COIN FLIPPING (CONTINUED)

TEACHER MOVES

Students may make errors during the Exploration, including forgetting or missing some possible outcomes in the sample space. Allow the opportunity for them to correct their own thinking as they work through with their partner and compare with another partner pair in question 4d. Part of what students may realize by the end of the Exploration as they move into the 14.1.4 Guided Instruction is that using an organization tool like a tree diagram can be helpful in determining the sample space for repeated experiments. Do not be afraid to let the mistakes happen, just be sure to discuss them during the transition to the next component.

ENRICHMENT

Encourage students to utilize the table to predict the sample space of getting all heads for four coins. Use intentional questioning to help frame student thinking, if needed:

- What do you notice about the number of outcomes as the number of coins increases by 1?
- What do you notice about the likelihood of getting heads as the number of coins increases? How can you use this to inform your prediction?

14.1.4 GUIDED INSTRUCTION: TREE DIAGRAMS

TEACHER MOVES

This component begins with the same repeated experiment as the exploration to focus on the benefits of using a tree diagram to visualize the sample space prior to introducing a new repeated experiment. It is helpful to point out the labels “First Toss” and “Second Toss,” as these are not typically on a tree diagram but are placed here to provide understanding. Draw attention to the fact that the Hs in the second toss represent different outcomes. Since this is a repeated experiment, the outcomes include both the result in the first toss and second toss. Creating a tree diagram provides visual entry point to visualize the sample space.

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

The intent is for students to visualize the random process with tree diagrams. To support MA.K12.MTR.2.1, encourage students to compare the two representations – tree diagrams versus lists – and how they can each be useful representations for different repeated experiments.



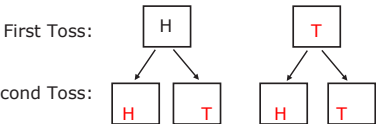
14.1.4 Guided Instruction: Tree Diagrams

In addition to making a list, tree diagrams can also be used to determine the sample space for a repeated experiment.



This display is called a **tree diagram**, and it helps visualize the sample space.

- Follow the prompts below to complete a tree diagram showing all possible outcomes when a fair coin is flipped two times.

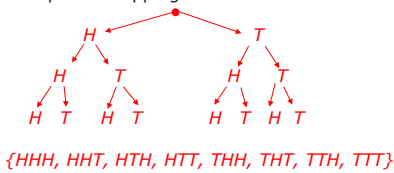


Sample Space: {HH, **HT**, **TH**, **TT**}

- Write the possible outcomes for the first coin toss.
H, T
- Write the possible outcomes for the second coin toss.
H, T
- Write the sample space.
{HH, HT, TH, TT}
- Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Flipping a fair coin two times	2	2	4

2. Suppose a fair coin is tossed three times.
- a. Create a tree diagram to determine the sample space of flipping a fair coin three times.



- b. How many outcomes are in the sample space?

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Number of Outcomes in the Third Step	Total Number of Outcomes in the Sample Space
Flipping a fair coin three times	2	2	2	8

- c. How many outcomes are in Event A = Flipping all heads?
1
- d. How many outcomes are in Event B = Flipping exactly one head?
3

Tree diagrams can also be set up horizontally.

14.1.4 GUIDED INSTRUCTION: TREE DIAGRAMS (CONTINUED)

TEACHER MOVES

Students may not identify the tree diagram as a sample space, and instead think that they need to list the sample space after creating a tree diagram. Provide an opportunity for students to make connections between the definition of sample space and the tree diagram as a visual display of all outcomes. Have students also list the sample space in set notation after creating the tree diagram. This response could be coupled with the 14.1.5 Wrap-Up for you to evaluate student progress from the lesson, as well as to provide some clear understanding on how both the list and tree diagram fully represent the sample space, but one may be preferable to the other.

ENRICHMENT

Which event is more likely – Event A or Event B? How do you know which is more likely based on the sample space?

14.1.4 GUIDED INSTRUCTION: TREE DIAGRAMS (CONTINUED)

TEACHER MOVES

Think-Pair-Share

After introducing the new context, consider having students work independently to complete the tree diagram for rolling a 3 on the first roll. Students then pair up and compare their answers. Prompt students to discuss either of the following:

- As we only have part of the tree diagram, how can we predict the total number of outcomes in this experiment?
- Tree diagrams can be used to determine the sample space for repeated experiments with more than two steps. Brainstorm some repeated experiments (other than tossing a coin or rolling a die) that can be represented using a tree diagram.

Listen as students discuss their ideas. Then, have some students share their thoughts with the class.

3. The sample space for rolling a fair die two times can be displayed in a tree diagram. Complete the tree diagram for rolling a 3 on the first roll.

First Roll	Second Roll	Outcome
1	1	1, 1
	2	1, 2
	3	1, 3
	4	1, 4
	5	1, 5
	6	1, 6
2	1	2, 1
	2	2, 2
	3	2, 3
	4	2, 4
	5	2, 5
	6	2, 6
3	1	3, 1
	2	3, 2
	3	3, 3
	4	3, 4
	5	3, 5
	6	3, 6

This tree diagram only shows half of the sample space because it does not include the outcomes for rolling a 4, 5, or 6 on the first roll, yet it fills at least half a page. Even though a tree diagram can take up a lot of space, they are especially useful because they can be used to determine the sample space for repeated experiments with any number of steps, not just two.



14.1.5 Wrap-Up: Reflection

1. Use the scale below to rate your ability to determine the sample space of a random process.



0 = I don't know what a sample space is.

10 = I can teach someone else how to determine the sample space of a random process.

Answers will vary.

2. What questions or comments do you have about determining the sample space of a random process?

Answers will vary.

14.1.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons. Consider having students specify whether they struggled with listing the sample space or with creating tree diagrams.

QUESTIONING

Couple student responses here with question 2 in the 14.1.4 Guided Instruction to evaluate student progress from the lesson.